

FraudShield

A Multi-Modal Fraud Risk Scoring Engine for Banking Applications

Project Abstract • Team FraudShield • Hackathon Idea Submission • 2026

Abstract

Financial institutions increasingly face fraud that exploits generative AI — deepfake video during KYC onboarding, cloned voices in vishing/OTP scams, and automated bots mimicking genuine user sessions. Most verification systems still rely on a single signal, such as a face-liveness check or a password, which modern synthetic-media techniques can now defeat in isolation. We present **FraudShield**, a multi-modal fraud risk scoring engine that fuses three independent signals — face liveness, voice liveness, and device/network behavior — into one explainable risk score, evaluated at the moment of a sensitive banking action such as login, video KYC, or a high-value transfer.

FraudShield comprises four components. A **face module** analyzes short video clips using MediaPipe Face Mesh to compute blink rate and frame-to-frame landmark jitter, two liveness cues that deepfake and face-swap renders struggle to reproduce naturally. A **voice module** extracts pitch-contour variance, pause ratio, spectral flatness, and MFCC delta irregularity from audio using Librosa to flag synthetic or cloned speech. A **device/network module** trains an unsupervised IsolationForest over session telemetry — device-fingerprint novelty, IP/ASN risk, login hour, and behavioral biometrics such as mouse-movement entropy and typing cadence — to flag anomalous sessions without requiring labeled fraud data. A **fusion engine** combines the three module scores into a single 0–100 fraud score and a LOW/MEDIUM/HIGH RISK verdict, using a justified weighted sum by default with a documented upgrade path to a learned logistic-regression stacker once labelled outcomes are available. Every score is accompanied by human-readable reasons, making the system explainable to compliance and fraud-operations teams rather than a black box.

The MVP is implemented in Python with OpenCV, MediaPipe, Librosa, and scikit-learn, and exposed through a Streamlit dashboard that lets a user simulate a session, run a fraud check, and inspect the score breakdown, reasons, and the JSON response a banking backend would receive. End-to-end testing confirms the system correctly distinguishes a genuine session (risk 7/100) from a fraudulent one exhibiting a new device, foreign IP, high-risk network, unusual login hour, and robotic interaction patterns (risk 73–79/100, HIGH RISK), with fusion weights renormalized so a missing signal is never mistaken for a safe one. Each heuristic module has a clear, low-effort upgrade path to a pretrained deep model (FaceForensics++/Celeb-DF checkpoints for face, ASVspoof/RawNet2 for voice) without altering the surrounding architecture, making FraudShield a practical, explainable, and extensible foundation for real-time fraud prevention in digital banking.

Keywords: *fraud detection, deepfake detection, voice-clone detection, liveness detection, anomaly detection, multi-modal fusion, explainable AI, digital banking, KYC security*

System at a Glance

Component	Signal	Core Technique
Module A – Face	Deepfake / liveness	MediaPipe Face Mesh: blink rate + landmark jitter
Module B – Voice	Voice-clone detection	Librosa: pitch variance, pause ratio, spectral flatness, MFCC delta
Module C – Device	Session anomaly	scikit-learn IsolationForest + rule-based fingerprint/behavioral checks
Module D – Fusion	Combined verdict	Weighted sum (MVP) → logistic-regression stacker (stretch)